



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

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Artificial Intelligence

## **Development and Analysis of a Superstore Sales Dashboard Using Power BI for Business Intelligence Insights**

### **ASSIGNMENT :**

**POWER BI VISUALIZATION – (Superstore Sales Dashboard)**

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# 1. Introduction

In today's data-driven business environment, organizations increasingly rely on data analytics and business intelligence (BI) tools to support strategic decision-making. Sales data, in particular, plays a crucial role in understanding customer behavior, product performance, and market trends. However, raw transactional data alone does not provide meaningful insights unless it is systematically processed, analyzed, and visualized (Sharda, Delen and Turban, 2020).

Many organizations face challenges in extracting actionable insights from large volumes of sales data due to fragmented datasets, lack of proper data modeling, and limited visualization capabilities. Without effective analytical tools, decision-makers may struggle to identify key performance drivers, detect inefficiencies, or respond to market changes in a timely manner. This gap highlights the importance of integrating data analytics with visualization tools to transform raw data into meaningful business insights.

To address this challenge, this project focuses on developing an interactive sales dashboard using Power BI based on the Superstore dataset. Power BI is a widely used business intelligence tool that enables users to perform data modeling, create interactive visualizations, and derive insights through data exploration (Microsoft, 2023). By leveraging Power BI, this project aims to present sales data in a structured and visually intuitive manner, allowing users to analyze trends, compare performance across dimensions, and identify key areas for improvement.

The primary purpose of this project is to design and implement a comprehensive dashboard that supports descriptive and diagnostic analysis of sales performance. The dashboard provides insights into sales trends over time, profitability across product categories, regional performance variations, and customer segmentation. Through these analyses, the project demonstrates how business intelligence techniques can enhance decision-making by providing clear, data-driven insights.

## 2. Objective

The specific objectives are as follows:

- To transform raw sales data into a structured data model using appropriate data preparation and modeling techniques.
- To design and develop interactive dashboards that effectively visualize key business metrics such as sales, profit, orders and customer activity.
- To analyze sales performance across different dimensions, including time, product categories & sub-categories, regions & states, and customer segments.
- To identify patterns, trends, and anomalies in the data that can support better business decisions.
- To demonstrate the application of business intelligence tools and techniques in solving real-world analytical problems.

## 3. Scope

This study uses the Superstore Sales dataset obtained from Kaggle, containing approximately **9,994 rows** of transactional data.

Dataset link: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

The dataset includes attributes such as **Row ID, Order ID, Order Date, Ship Date, Ship mode, Customer ID, Customer Name, Segment, Product ID, Category, Sub-Category, Product Name, Sales, Profit, Quantity, Discount, Region, State, City and Country** enabling analysis of sales performance, customer behavior, and regional trends. Microsoft Power BI Desktop is used for data cleaning, transformation, modeling, and visualization. A star schema is implemented to organize the data efficiently. The dashboard focuses on descriptive and diagnostic analysis, using visuals such as KPI cards, bar charts, line charts, scatter plots, and tables to present insights. This study is limited to a single dataset and does not include external factors such as market conditions or competitor data. Additionally, the analysis is limited to descriptive and diagnostic insights without predictive modeling.

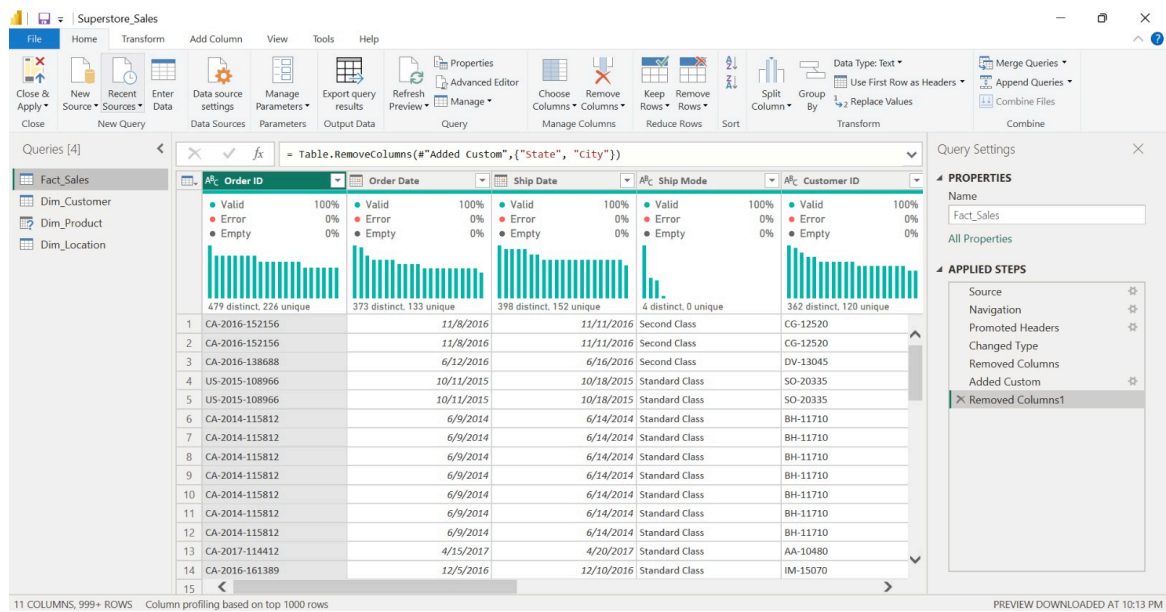
## 4. Dataset

The dataset used in this study is the Superstore Sales dataset obtained from Kaggle, which contains approximately 9,994 rows of transactional data representing retail operations. This dataset is particularly suitable for business intelligence analysis as it includes comprehensive attributes such as **Order ID, Order Date, Customer ID, Customer Segment, Customer Name, Product ID, Category, Sub-Category, Sales, Profit, Quantity, Discount, and Region**. These attributes allow for multi-dimensional analysis across time, products, customers, and geographical locations, enabling the identification of sales trends, profitability patterns, and customer behavior. The dataset is structured and sufficiently rich to support the development of a star schema model, making it highly appropriate for Power BI-based data modeling and visualization. Its relevance to the study lies in its ability to simulate real-world retail scenarios, thereby providing meaningful insights into business performance and supporting data-driven decision-making.

## 5. Step of processing

### 5.1 Prepare, Load, and Transform.

The data processing stage involves three key phases: preparation, loading, and transformation, ensuring the dataset is clean, structured, and suitable for analysis. Initially, the Superstore dataset was prepared by checking for data consistency, correcting data types (e.g., converting dates and numeric fields), removing unneeded columns such as **Row ID** and ensuring no critical missing or duplicate values that could affect analysis. The cleaned dataset was then loaded into Microsoft Power BI Desktop, where it was organized for further processing. In the transformation phase, Power Query was used to structure the data into a star schema model by separating it into a central fact table (**Fact\_Sales**) and multiple dimension tables (**Dim\_Customer, Dim\_Product, Dim\_Location, and Dim\_Date**). Relationships were established between these tables using primary and foreign keys to enable efficient data integration. Additional transformations included creating calculated columns (e.g., **Total Cost**) and **DAX measures** (e.g., **Total Sales, Profit Margin**) to support analysis. This structured approach ensures that the data model is optimized for performance and allows for accurate, flexible, and interactive visualizations within the dashboard. Figure 1 illustrates the data preparation and transformation process within Power BI, including data cleaning, table structuring, and relationship modeling.



**Figure 1: Data preparation and transformation process in Power BI Power Query Editor**

## 5.2 Data Model

The data modeling and integration process was implemented using a **star schema**, where a central fact table is connected to multiple dimension tables. As shown in **Figure 2**, the Fact\_Sales table contains transactional data such as Sales, Profit, Quantity, and Discount, and is linked to four dimension tables: Dim\_Customer, Dim\_Product, Dim\_Date, and Dim\_Location, which provide contextual information for analysis.

Each relationship follows a **one-to-many (1:\*) structure**, where the dimension tables represent unique entities and the Fact\_Sales table contains multiple related transactions. For example, Dim\_Customer links customer information to multiple sales records, while Dim\_Product enables analysis by category and sub-category. The Dim\_Date table supports time-based analysis such as monthly and yearly trends. Additionally, the Dim\_Location table provides geographical context, linking attributes such as City and Region to the fact table, allowing analysis of sales and profit across different regions.

The integration flow begins from the dimension tables, which filter and provide context to the Fact\_Sales table. For instance, selecting a specific category in Dim\_Product filters the related records in Fact\_Sales, which then updates all associated measures such as Total Sales and Profit. This directional flow ensures efficient querying and supports interactive analysis within the dashboard. This structured model improves query performance, reduces redundancy, and supports efficient and interactive analysis within the dashboard. Figure 2 illustrates the star schema data model and relationships between fact and dimension tables in Power BI.

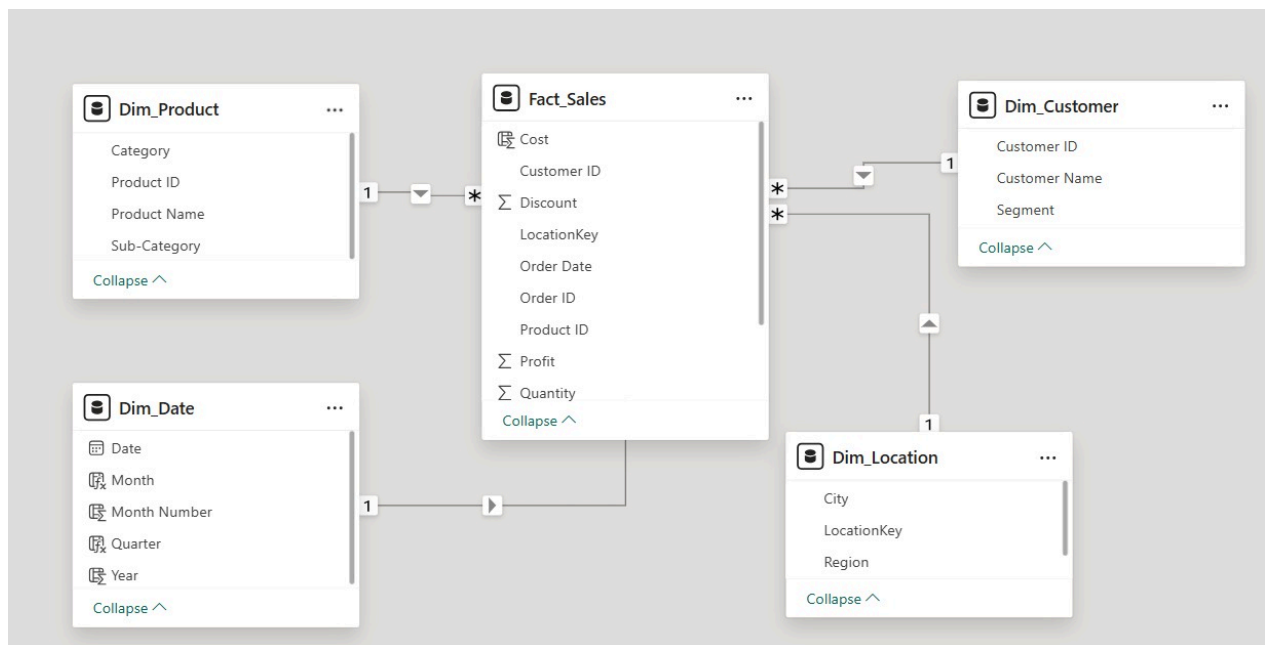


Figure 2: Star Schema Data Model

## 5.3 Visualization Report and Dashboard

### 5.3.1 Executive Summary Dashboard

Figure 3 presents the Executive Summary dashboard, which provides a comprehensive overview of key business performance indicators. KPI cards are used to display essential metrics, including Total Sales (\$2.30M), Total Profit (\$286.40K), Total Orders (5K), Total Customers (793), and Profit Margin (12.47%). These indicators allow users to quickly assess overall business performance at a glance.

A **line chart** is used to illustrate the trend of Total Sales and Total Profit from 2014 to 2017. This visualization shows a consistent upward trend, indicating business growth over time, with a noticeable increase in both sales and profit in the later years.

Several **bar charts** are included to provide comparative analysis across different dimensions. The “Sales by Category” chart highlights that Technology generates the highest sales, followed by Furniture and Office Supplies. The “Profit by Region” chart shows that the West region contributes the highest profit, while Central has the lowest, indicating regional performance differences. Additionally, the “Sales by Customer Segment” chart reveals that the Consumer segment generates the highest revenue compared to Corporate and Home Office segments.

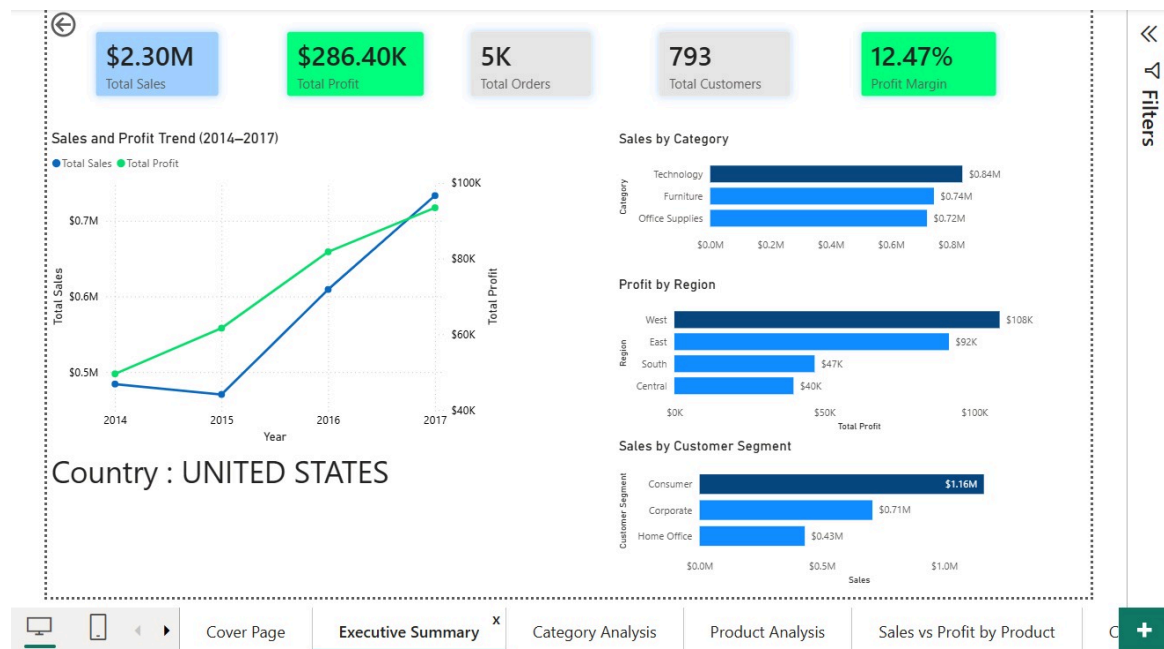


Figure 3 : Executive Summary Dashboard

### 5.3.2 Category Analysis Dashboard

Figure 4 presents the Category Analysis dashboard, which provides a more detailed examination of sales and profitability at the category and sub-category levels. A **slider (filter)** is included to allow users to select specific product categories (Furniture, Office Supplies, and Technology), enabling dynamic and focused analysis.

At the top, **KPI cards** display key performance metrics such as Total Sales, Total Profit, Total Orders, and Profit Margin, providing a quick summary of performance based on the selected category.

A **line chart** titled “Category Sales Trend (2014–2017)” is used to show the category sales trend over time. The visualization indicates a steady increase in sales, particularly from 2015 onwards, suggesting consistent growth within the selected category.

Several **bar charts** are used to provide comparative insights across sub-categories. The “Sales by Sub-Category” chart identifies top-performing sub-categories, such as Phones (\$0.3M) and Chairs (\$0.3M), which generate the highest sales. The “Orders by Sub-Category” chart highlights demand patterns, showing that items like Binders (1316) and Paper (1191) have the highest number of orders. Additionally, the “Profitability by Sub-Category” chart illustrates profit performance, where sub-categories such as Copiers (\$56K) and Phones (\$45K) contribute the highest profit, while some sub-categories show lower profitability.



Figure 4: Category Analysis Dashboard

### 5.3.3 Product Analysis Dashboard

Figure 5 presents the Product Analysis dashboard, which provides a detailed evaluation of product-level performance across sales, profit, and order volume. Multiple **slicers** are included for Year, Category, and Sub-Category, allowing users to dynamically filter and analyze specific segments of the data.

At the top, **KPI cards** display key metrics such as Total Sales, Total Profit, Total Orders, and Profit Margin, providing an overall performance summary based on selected filters.

Several **bar charts** are used to compare product performance. The “Products by Sales” chart highlights the top-selling products, with items such as the Canon imageCLASS series generating the highest sales (\$62k). The “Profit by Product” chart identifies the most profitable products, which is also Canon imageCLASS series (\$25.02K), but not all high sales products have high profit. For Example Cisco TelePresence is at third position in top sales but has low profit compared to others. Additionally, the “Orders by Product” chart provides insights into product demand, indicating frequently purchased items such as staple-envelope products around 48.

A **table visual** titled “Top 10 Products” is included to present detailed information, including product name, sales, profit, and number of orders. This allows users to examine exact values and identify products with strong or weak performance, including those with low or negative profit despite high sales.

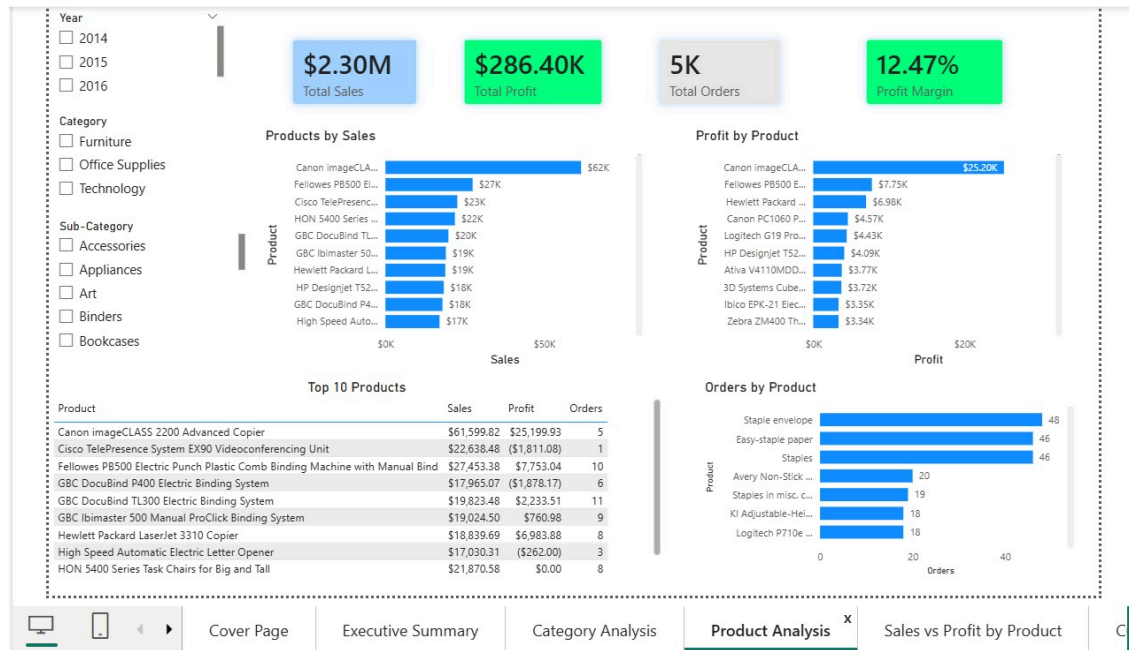


Figure 5: Product Analysis Dashboard

### 5.3.4 Sales vs Profit by Product Dashboard

Figure 6 presents the Sales vs Profit by Product dashboard, which provides a diagnostic analysis of the relationship between sales and profit at the product level. A **scatter plot** is used as the main visualization, where each point represents an individual product. The horizontal axis (X-axis) shows Total Sales, while the vertical axis (Y-axis) represents Total Profit, allowing for the identification of patterns and relationships between these two variables.

The scatter plot reveals that products are distributed across different performance levels. Products located in the upper-right region of the chart represent high sales and high profit, indicating strong-performing products such as Canon imageCLASS series. In contrast, products with high sales but low or negative profit appear in the lower-right area, highlighting potential inefficiencies such as high costs or excessive discounts. Additionally, products with low sales and low profit are clustered near the origin, indicating lower-performing items.

**Slicers** for Year, Category, and Sub-Category are included to enable interactive filtering, allowing users to analyze specific segments and identify trends within particular groups of products.



Figure 6: Sales vs Profit by Product Dashboard

### 5.3.5 Customer Segment Dashboard

Figure 7 presents the Customer Segment Analysis dashboard, which examines sales performance across different customer segments. A **slicer** for Region is included, allowing users to filter and analyze segment performance within specific geographical areas.

At the top, **KPI cards** display key metrics such as Total Sales, Total Profit, Total Customers, and Profit Margin, providing a quick overview of performance based on the selected region.

The dashboard includes multiple **bar charts** to compare customer segments. The “Sales by Customer Segment” chart shows that the Consumer segment generates the highest sales (\$1.16M), followed by Corporate (\$0.7M) and Home Office segments (\$0.43M). Similarly, the “Orders by Customer Segment” chart indicates that the Consumer segment has the highest number of orders (2.6K), reflecting stronger customer activity and demand.

The “Profit by Segment” chart further reveals that the Consumer segment contributes the highest profit (\$0.13M), while the Home Office segment generates the lowest profit (\$0.06M). This indicates that the Consumer segment is the most valuable segment in terms of both revenue and profitability.

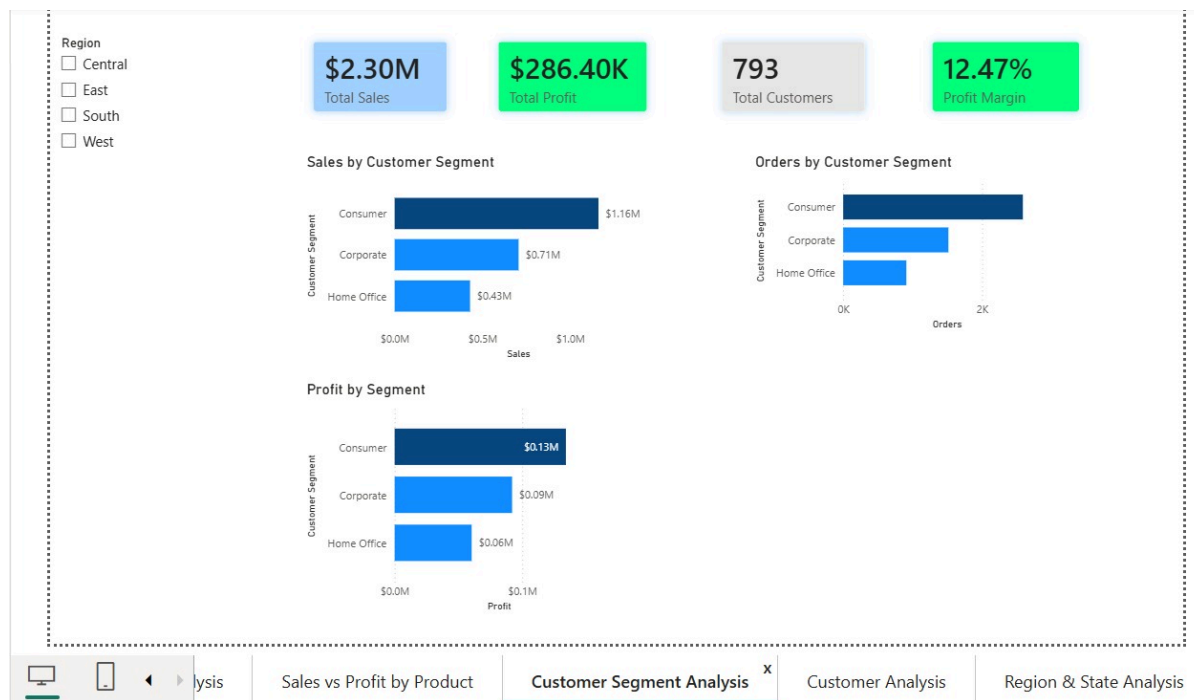


Figure 7: Customer Segment Analysis Dashboard

### 5.3.6 Customer Analysis Dashboard

Figure 8 presents the Customer Analysis dashboard, which provides a detailed evaluation of individual customer performance in terms of sales, profit, and order activity. **Slicers** for Segment and Region are included, enabling users to filter and analyze customer behavior across different groups and geographical areas.

At the top, **KPI cards** display key metrics such as Total Sales, Total Profit, Total Customers, and Profit Margin, offering a quick overview of overall customer-related performance.

Several **bar charts** are used to highlight top-performing customers. The “Top 10 Customers by Sales” chart identifies customers such as Sean Miller as the highest contributors to revenue. The “Top 10 Customers by Order” chart shows customers with the highest number of transactions, such as Emily Phan, indicating frequent purchasing behavior. Additionally, the “Profitability by Customer” chart highlights customers who generate the highest profit, such as Tamara Chand, emphasizing that the most profitable customers are not always the highest in sales or order count.

A **table visual** titled “Top 5 Customers by Profits” provides detailed information, including customer name, segment, sales, profit, and total orders. This allows for precise comparison and deeper evaluation of customer value.

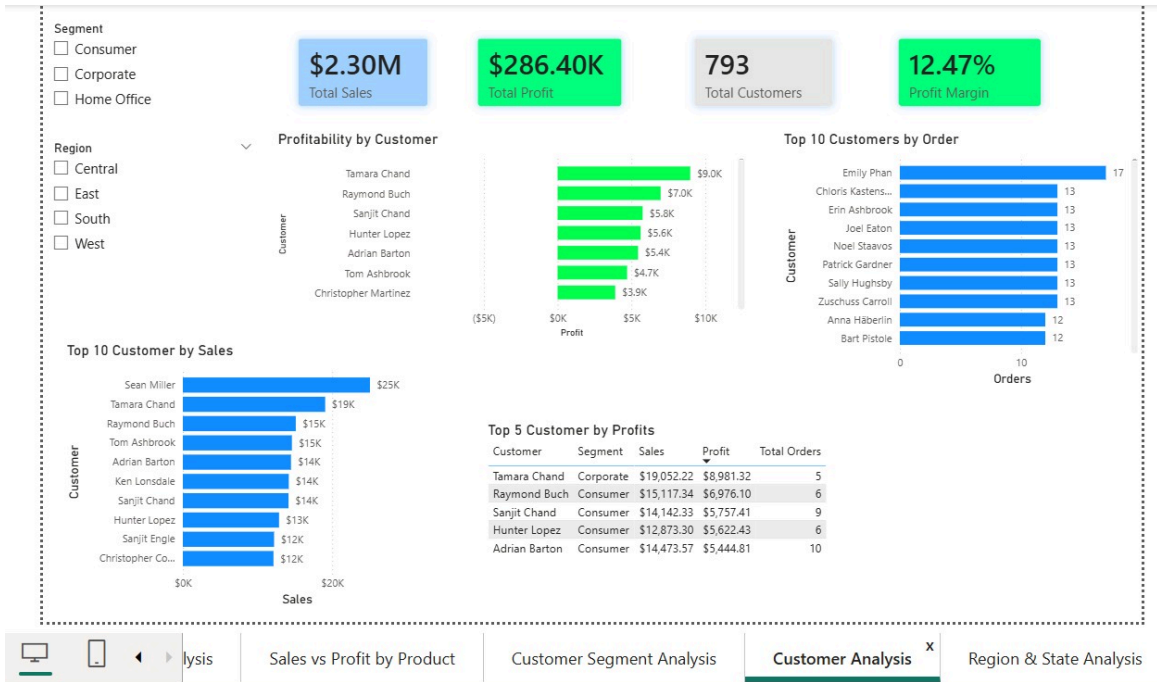


Figure 8: Customer Analysis Dashboard

5.3.7 Region & State Analysis Dashboard

Figure 9 presents the Region and State Analysis dashboard, which provides a detailed comparison of sales, profit, and order performance across different geographical locations. At the top, **KPI cards** summarize key metrics, including Total Sales, Total Profit, Total Customers, Total Orders, and Profit Margin, offering a quick overview of overall performance.

The dashboard includes multiple **bar charts** to compare performance across regions. The “Sales by Region” chart shows that the West region generates the highest sales, followed by East, Central, and South. Similarly, the “Profit by Region” chart indicates that the West region contributes the highest profit, while Central and South regions show relatively lower profitability. The “Orders by Region” chart further confirms that the West region has the highest order volume, suggesting strong market activity.

At a more detailed level, the dashboard presents **state-level analysis** through additional bar charts. The “Sales by State” chart highlights California and New York as the top-performing states in terms of revenue. The “Profit by State” chart shows that California generates the highest profit, followed by New York and Washington. The “Orders by State” chart indicates that California has the highest number of orders, reflecting strong customer demand in that state.

Overall, this dashboard effectively uses KPI cards and bar charts to provide both regional and state-level insights. It enables users to identify high-performing geographical areas, understand variations in sales and profitability, and support location-based business strategies and decision-making.

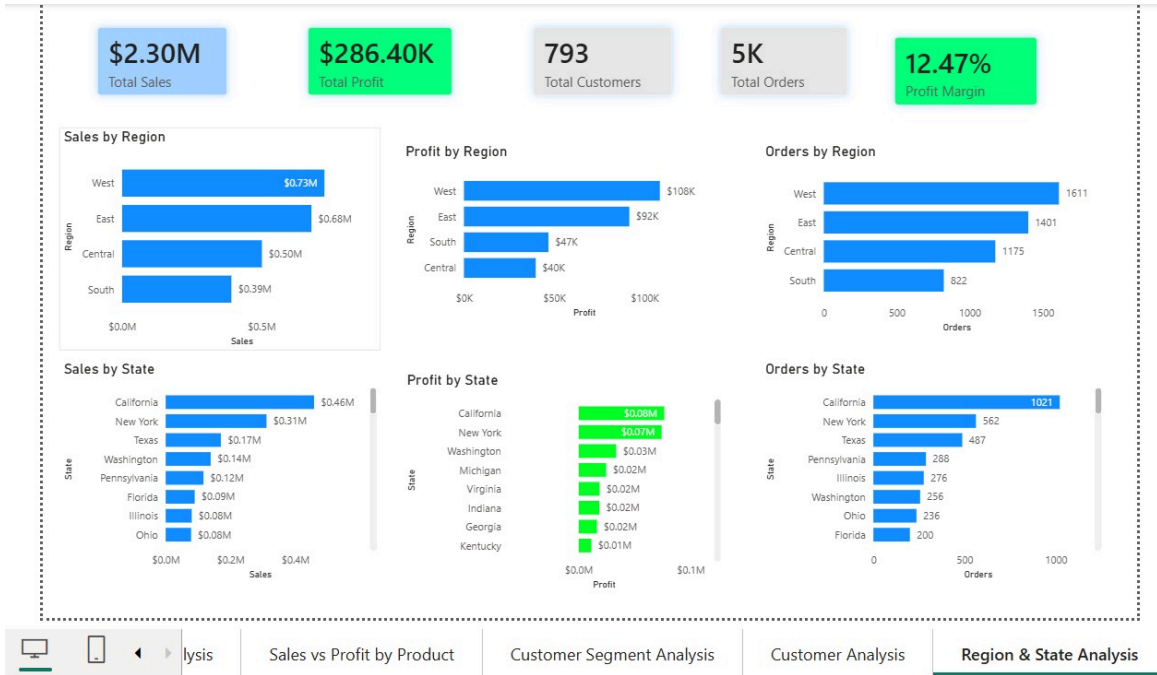


Figure 9: Region & State Analysis Dashboard

## 6. Result Finding and Discussion

The developed dashboard provides comprehensive insights into sales performance, profitability, customer behavior, and regional distribution. The findings are derived from multiple visualizations, enabling both descriptive and diagnostic analysis to support data-driven decision-making.

### 6.1 Overall Business Performance

As shown in **Figure 3 (Executive Summary Dashboard)**, the business achieved total sales of approximately \$2.30M and total profit of \$286.40K, resulting in a profit margin of 12.47%. The sales and profit trend over time indicates consistent growth from 2014 to 2017, demonstrating an overall positive business trajectory. This suggests that the company has been successful in increasing revenue while maintaining profitability.

However, the moderate profit margin indicates that while sales are growing, there may be underlying cost or discount factors affecting overall profitability, requiring further investigation.

### 6.2 Sales and Profit Trends Over Time

The line chart in **Figure 3** shows that both sales and profit increase steadily over the years, with a significant rise after 2015. This indicates improved business performance and possibly better market penetration or product demand.

From a critical perspective, although profit follows a similar upward trend as sales, the growth rate is not proportionally higher. This suggests that increased sales do not necessarily translate into equally higher profit, highlighting potential inefficiencies such as increased costs or high discount usage.

### 6.3 Category and Sub-Category Performance

Based on **Figure 3 (Executive Summary Dashboard)**, Technology is identified as the top-performing category in terms of sales, while based on **Figure 4 (Category Analysis Dashboard)** sub-categories such as Phones and Chairs generate the highest sales each \$0.33M. Additionally, items like Binders (1316) and Paper (1191) show high order volumes, indicating strong demand.

However, profitability analysis reveals variations across sub-categories. Some products with high sales do not generate equally high profit, indicating that certain items may be less efficient or impacted by discounts and operational costs. This highlights the importance of not relying solely on sales volume as a performance indicator.

## 6.4 Product-Level Insights

From **Figure 5 (Product Analysis Dashboard)**, specific products such as the Canon imageCLASS series contribute significantly to both sales and profit. However, the dashboard also reveals that some products generate high sales but relatively low or zero profit.

This finding is further supported by **Figure 6 (Sales vs Profit by Product Dashboard)**, where the scatter plot clearly shows that some products with high sales fall into low or negative profit regions. This indicates inefficiencies, possibly due to high discounts, high costs, or pricing issues.

This diagnostic insight is critical, as it suggests that the company should focus on optimizing pricing strategies, reducing unnecessary discounts, or controlling costs for such products to improve profitability.

## 6.5 Customer Segment Analysis

As illustrated in **Figure 7 (Customer Segment Analysis Dashboard)**, the Consumer segment generates the highest sales, orders, and profit, making it the most valuable customer group. The Corporate segment follows, while the Home Office segment contributes the least.

This indicates that the business is heavily dependent on individual consumers. From a strategic perspective, this suggests an opportunity to further expand into corporate or home office markets to diversify revenue sources and reduce dependency on a single segment.

## 6.6 Customer-Level Insights

From **Figure 8 (Customer Analysis Dashboard)**, it is evident that top customers differ across sales, profit, and order frequency. For example, customers with the highest sales are not necessarily the most profitable, and those with the highest number of orders do not always contribute the most revenue.

This highlights the importance of evaluating customer value from multiple perspectives rather than relying on a single metric. Businesses can use this insight to identify high-value customers and develop targeted marketing strategies, such as loyalty programs or personalized offers.

## 6.7 Regional and State Performance

Based on **Figure 9 (Region and State Analysis Dashboard)**, the West region consistently performs the best in terms of sales, profit, and order volume, followed by the East region. At the state level, California and New York are identified as the top contributors to both revenue and profit.

In contrast, regions such as Central and South show relatively lower performance. This indicates geographical disparities in business performance, suggesting that the company may need to investigate market conditions, logistics, or customer demand in lower-performing regions.

## 6.8 Impact of Discounts on Profitability

The analysis across product and category levels suggests that high discounts may negatively impact profitability. As observed in the **scatter plot figure** in the Sales vs Profit by Product Dashboard, some products achieve high sales but generate low or negative profit, indicating that excessive discounting may reduce profit margins.

This highlights the need for better discount management strategies to balance sales growth with profitability.

## 6.9 Business Implications and Recommendations

Based on the findings, several key business implications can be derived:

- The company should focus on optimizing pricing and discount strategies to improve profit margins.
- High-performing categories such as Technology should be further promoted, while underperforming sub-categories should be reviewed.
- Expansion strategies should target less dominant customer segments and regions to diversify revenue sources.
- Product-level analysis should be used to identify and improve low-profit items despite high sales.
- Customer segmentation strategies should prioritize high-value customers for retention and growth.

## 7. Conclusion

This study successfully developed an interactive Power BI dashboard to analyze the Superstore dataset and generate meaningful business insights. The findings reveal that while the business demonstrates steady growth in sales and profit over time, profitability is not consistently aligned with sales performance, particularly at the product and sub-category levels.

Key insights highlight that the Technology category and the Consumer segment are the main contributors to revenue and profit, while certain products generate high sales but low or negative profit due to factors such as discounting. Additionally, regional analysis shows that the West region and states such as California lead in overall performance, whereas other regions require improvement.

Overall, the dashboard effectively transforms raw data into actionable insights, supporting better decision-making in areas such as pricing, product strategy, and customer targeting. This project demonstrates the practical value of business intelligence tools in enhancing data-driven decision-making and provides a strong foundation for further analytical development.

## References

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